Rishabh Mudgal

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Performance-driven AI and Python Backend Engineer with strong expertise in integrating Large Language Models (LLMs), Retrieval-Augmented Generation (RAG) pipelines, and orchestrating scalable web APIs (FastAPI, Flask, Django). Proven track record in designing production-grade ETL architectures, deploying agentic workflows, and optimizing system resource utilization (GPU/CPU) to deliver high-availability, multimodal AI applications and automated data pipelines.

SKILLS

- **AI, LLM & Data Pipelines:** Retrieval-Augmented Generation (RAG), Agentic Workflows, Prompt Engineering, LLMs (Ollama, DeepSeek, Gemini, Groq, Qwen2.5), Text-to-Speech (Kokoro-ONNX), Computer Vision (OpenCV, Stable Diffusion, ControlNet), PyTorch, Hugging Face, SuryaOCR.
- **Backend & API Development:** Python, FastAPI, Flask, Django, RESTful APIs, Node.js, Asyncio, MVC Architecture.
- **Databases & Caching:** PostgreSQL, MySQL, MongoDB, Redis, SQL Optimization.
- **Data Engineering:** Pandas, NumPy, ETL Pipelines, Web Scraping (Selenium, BeautifulSoup, ScrapingAnt API), CRON Workflow Automation.
- **DevOps & Infrastructure:** Docker, AWS (EC2, S3), RunPod GPU Orchestration, Linux, Git, CI/CD pipelines.

WORK EXPERIENCE

Software Engineer | Pentation Analytics
Jun 2024 - Present

- Designed, developed, and optimized robust Python REST APIs using Flask and FastAPI for automated data ingestion, overhauling backend architecture to reduce endpoint response and execution times by 90%.
- Engineered a hybrid LLM-driven OCR backend architecture utilizing SuryaOCR on RunPod GPU instances; dynamically routed payloads via Gemini and Groq APIs for high-accuracy, low-latency multimodal data extraction.
- Deployed enterprise-grade Python ETL pipelines (Selenium, BeautifulSoup, Pandas, SQL) to ingest, normalize, and process large-scale unstructured legal datasets for insurance fraud detection platforms.
- Applied advanced prompt engineering and JSON-constrained formatting to maximize OCR-to-LLM extraction consistency across highly unstructured enterprise PDFs.
- Automated end-to-end data workflows and backend processes via CRON scheduling, guaranteeing reliable, near real-time processing under strict SLA timelines.
- Partnered with frontend engineers following Agile methodologies to integrate analytical backend APIs with enterprise dashboards, driving a 25% increase in operational decision-making efficiency.

Full-Stack Developer | Tutcart
Dec 2023 - Apr 2024

- Developed a scalable Node.js backend comprising 17+ microservice REST APIs, implementing strict status codes, comprehensive error-handling middleware, and token-based security parameters.
- Built a responsive Next.js web application with enhanced security protocols, leading to a 15% boost in user engagement metrics.

Python Developer | MetroGhar Property Solutions
Oct 2023 - Dec 2023

- Consolidated 4 highly unstructured transactional datasets using Pandas and NumPy, executing data cleaning and optimization scripts that improved data integrity by 30%.

Data Analyst Intern | Conscript HR Advisors

Aug 2023 - Oct 2023

- Built distributed web scraping scripts using Selenium and requests to ingest 75K+ rows of recruitment market data into internal databases.

Backend Developer | Samishti Infotech
Jul 2023 - August 2023

- Built a real-time chat backend using Node.js, Socket.IO, and MongoDB, successfully deploying the cluster to AWS and Heroku environments.
- Optimized MongoDB aggregation pipelines and indexing strategies, increasing query execution performance by 35%.

PROJECTS

End-to-End AI Podcast Orchestrator | Spotify: linktr.ee/thenarrativenetwork

2026

Tech Stack: Python, PyTorch, Hugging Face (Qwen2.5-7B), Kokoro-ONNX (TTS), BeautifulSoup, Pydub

- Engineered a fully automated content extraction pipeline utilizing BeautifulSoup and Trafilatura to scrape, clean, and convert unstructured web/Wikipedia HTML into clean, structured Markdown datasets.
- Integrated a local Large Language Model (Qwen2.5-7B) via Hugging Face Transformers to programmatically rewrite factual articles into dynamic, cinematic audio scripts using precise prompt engineering and context chunking.
- Architected a local Text-to-Speech (TTS) audio engine utilizing Kokoro-ONNX and Pydub to synthesize multi-voice human-like audio tracks, dynamically overlay background music, and automate the final batch packaging of .wav episodes.

Medium Article Automated Summarizer | GitHub: github.com/mr-mudgal/medium_premium_scrapper

2025

Tech Stack: Python, FastAPI, Flask, Ollama (DeepSeek-R1 8B), Selenium, BeautifulSoup, CRON, Subprocesses, SMTP

- Built a production-grade Python backend automation pipeline and REST API architecture that ingests email payloads, extracts web content, and serves local AI insights automatically.
- Engineered a scalable, robust web scraping system using Selenium and BeautifulSoup to reliably extract premium and free unstructured article content.
- Developed a privacy-first AI summarization module using local LLMs (Ollama, DeepSeek-R1 8B) with prompt engineering and structured JSON outputs for accurate, non-lossy backend data processing.
- Automated end-to-end workflow orchestration and service lifecycle management using Python subprocesses and CRON scheduling to manage local AI application states.
- Implemented stateful deduplication, relational database logging, and automated email delivery backend systems, improving pipeline efficiency, reliability, and reproducibility without cloud dependencies.

Multimodal Generative AI Publishing Pipeline

2026

Tech Stack: Python, Django, Flask, PyTorch, Diffusers (SDXL, ControlNet), HuggingFace (Qwen LLM), OpenCV, ReportLab, System Optimization

- Architected an automated, end-to-end multimodal backend pipeline integrating Flask/Django microservices with local Large Language Models and vision models to programmatically generate, process, and compile structured PDF assets.

- Engineered strict JSON-constrained prompt logic for a local LLM (Qwen) to autonomously generate base image configurations, negative constraints, and dynamic color palettes without human intervention, ensuring deterministic API responses.
- Implemented computer vision workflows using OpenCV and ControlNet (Canny edge detection) to enforce strict spatial consistency across multi-stage image generation and automated SVG vectorization pipelines.
- Optimized GPU resource orchestration by developing automated Python utilities for VRAM flushing, garbage collection, and model CPU offloading, enabling the seamless sequential execution of text and heavy vision models in resource-constrained backend environments.

Social Media Automation

2024

Tech Stack: Python, Selenium, Colorama, Script Optimization, Automation Frameworks

- Developed reusable Python scripts using Selenium and Colorama to automate complex social media scheduling matrices, reducing manual operational effort by 60%.
- Implemented robust anti-bot bypass logic and exception handling routines in Python, ensuring stable script execution on restricted web platforms.
- Optimized backend automation workflows and string processing configurations, cutting transaction processing times by 40% through efficient scripting practices.

Amazon Scrapper | GitHub: github.com/mr-mudgal/Amazon-Scrapper

2023

Tech Stack: Python, Pandas, BeautifulSoup, ScrapingAnt API, Data Extraction

- Created a modular Python-based web scraper incorporating the ScrapingAnt API to cleanly bypass anti-scraping frameworks and securely extract structured product schemas.
- Streamlined data normalization and output delivery routines into structured formats using Python's Pandas module, enabling seamless downstream analysis of 10,000+ product records.
- Enhanced operational script efficiency and reduced API latency, lowering application runtime requirements by 30% through optimized network interactions.

CERTIFICATIONS

Python Programming – HackerRank

SQL Database Management – HackerRank

Data Science with Python – Great Learning

Data Visualization – Great Learning

Power BI – Great Learning

Selenium Automation – Google Developers Launchpad

EDUCATION

B. Tech. in Computer Science and Engineering

A.P.J. Abdul Kalam Technical University, Ghaziabad | 2024